TP3- Implement MST

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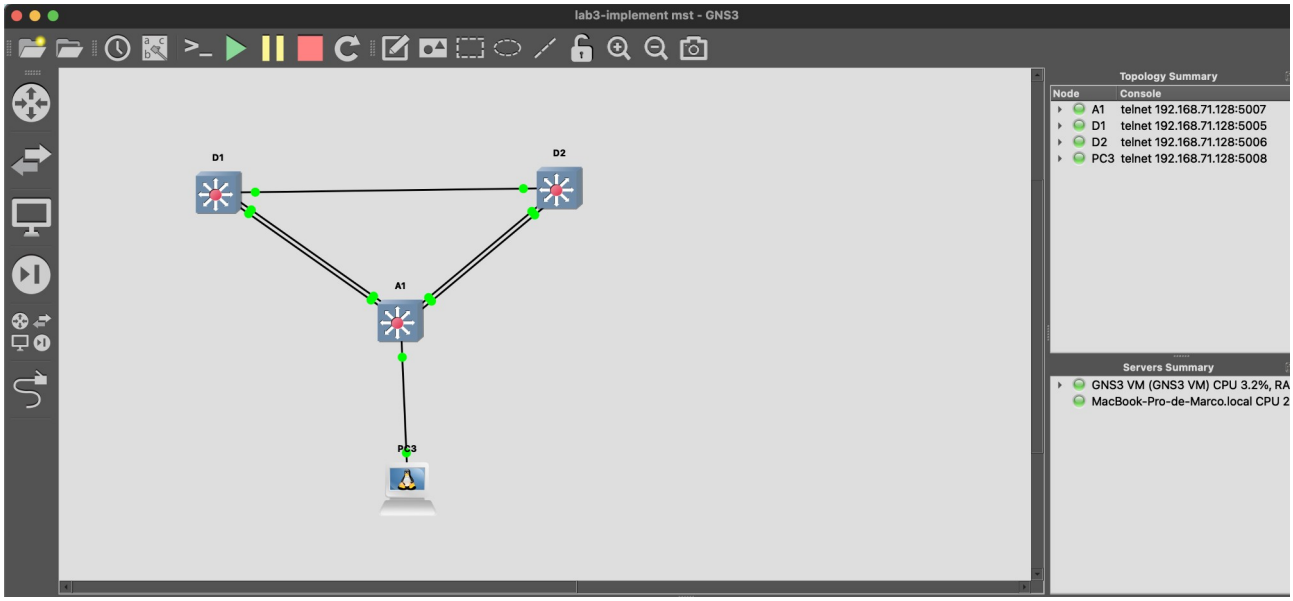
Topology



Bmarco11 — D1 — telnet 192.168.71.128 5005 — 80×47

```
*Oct 26 19:48:58.553: %LINK-5-CHANGED: Interface Ethernet1/3, changed state to a
dministratively down
*Oct 26 19:48:59.549: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to down
*Oct 26 19:48:59.549: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to down
*Oct 26 19:48:59.549: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to down
*Oct 26 19:48:59.554: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
*Oct 26 19:48:59.554: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
0, changed state to down
*Oct 26 19:48:59.554: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
1, changed state to down
*Oct 26 19:48:59.554: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
2, changed state to down
*Oct 26 19:48:59.554: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
3, changed state to down
D1(config-if-range)#exit
D1(config)#interface range e0/0,e0/1-2
D1(config-if-range)#switchport trunk encapsulation dot1q
D1(config-if-range)#switchport mode trunk
D1(config-if-range)#no shutdown
D1(config-if-range)#exit
*Oct 26 19:54:43.650: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Oct 26 19:54:43.651: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Oct 26 19:54:43.651: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to up
*Oct 26 19:54:44.654: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to up
*Oct 26 19:54:44.654: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
*Oct 26 19:54:44.654: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
D1(config-if-range)#exit
D1(config)#vlan 2
D1(config-vlan)#name secondvlan
D1(config-vlan)#exit
D1(config)#vlan 3
D1(config-vlan)#name thirdvlan
D1(config-vlan)#exit
D1(config)#vlan 4
D1(config-vlan)#name fourthvlan
D1(config-vlan)#exit
D1(config)#vlan 5
D1(config-vlan)#name fifthvlan
D1(config-vlan)#exit
D1(config)#
```

Bmarco11 — D2 — telnet 192.168.71.128 5006 — 80×48

```
dministratively down
*Oct 26 20:02:32.181: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to down
*Oct 26 20:02:32.181: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to down
*Oct 26 20:02:32.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to down
*Oct 26 20:02:32.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
D2(config-if-range)#exit
*Oct 26 20:02:32.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
0, changed state to down
*Oct 26 20:02:32.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
1, changed state to down
*Oct 26 20:02:32.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
2, changed state to down
*Oct 26 20:02:32.191: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
3, changed state to down
D2(config-if-range)#exit
D2(config)#interface range e0/0, e0/1-2
D2(config-if-range)#switchport trunk encapsulation dot1q
D2(config-if-range)#switchport mode trunk
D2(config-if-range)#no shutdown
D2(config-if-range)#
*Oct 26 20:04:12.946: %LINK-3-UPDOWN: Interface Ethernet0/0, changed state to up
*Oct 26 20:04:12.947: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Oct 26 20:04:12.947: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to up
*Oct 26 20:04:13.950: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
0, changed state to up
D2(config-if-range)#
*Oct 26 20:04:14.961: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
*Oct 26 20:04:14.962: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
D2(config-if-range)#exit
D2(config)#vlan 2
D2(config-vlan)#name secondvlan
D2(config-vlan)#exit
D2(config)#vlan 3
D2(config-vlan)#name thirdvlan
D2(config-vlan)#exit
D2(config)#vlan 4
D2(config-vlan)#name fourthvlan
D2(config-vlan)#exit
D2(config)#vlan 5
D2(config-vlan)#name fifthvlan
D2(config-vlan)#exit
D2(config)#
```

Bmarco11 — A1 — telnet 192.168.71.128 5007 — 80×47

```
*Oct 26 20:11:25.587: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
3, changed state to down
*Oct 26 20:11:25.588: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
0, changed state to down
*Oct 26 20:11:25.588: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
1, changed state to down
*Oct 26 20:11:25.588: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
2, changed state to down
*Oct 26 20:11:25.599: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
3, changed state to down
A1(config-if-range)#exit
A1(config)#interface range e0/1-2, e1/0-1
A1(config-if-range)#switchport trunk encapsulation dot1q
A1(config-if-range)#switchport mode trunk
A1(config-if-range)#no shutdown
A1(config-if-range)#
*Oct 26 20:13:58.218: %LINK-3-UPDOWN: Interface Ethernet0/1, changed state to up
*Oct 26 20:13:58.218: %LINK-3-UPDOWN: Interface Ethernet0/2, changed state to up
*Oct 26 20:13:58.219: %LINK-3-UPDOWN: Interface Ethernet1/0, changed state to up
*Oct 26 20:13:58.219: %LINK-3-UPDOWN: Interface Ethernet1/1, changed state to up
*Oct 26 20:13:59.223: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
1, changed state to up
*Oct 26 20:13:59.223: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet0/
2, changed state to up
A1(config-if-range)#
*Oct 26 20:14:00.231: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
0, changed state to up
*Oct 26 20:14:00.232: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/
1, changed state to up
A1(config-if-range)#exit
A1(config)#vlan 2
A1(config-vlan)#name secondvlan
A1(config-vlan)#exit
A1(config)#vlan 3
A1(config-vlan)#name thirdvlan
A1(config-vlan)#exit
A1(config)#vlan 4
A1(config-vlan)#name fourthvlan
A1(config-vlan)#exit
A1(config)#vlan 5
A1(config-vlan)#name fithvlan
A1(config-vlan)#name fifthvlan
A1(config-vlan)#exit
A1(config)#end
A1#
*Oct 26 20:16:31.947: %SYS-5-CONFIG_I: Configured from console by console
A1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5005 — 80×26

```
*Oct 26 20:19:18.963: %SYS-5-CONFIG_I: Configured from console by console
D1#show spanning-tree

MST0
  Spanning tree enabled protocol mstp
  Root ID    Priority    32768
             Address     aabb.cc00.0100
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32768  (priority 32768 sys-id-ext 0)
             Address     aabb.cc00.0100
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Et0/0               Desg FWD 2000000   128.1    Shr Bound(PVST)
Et0/1               Desg FWD 2000000   128.2    Shr Bound(PVST)
Et0/2               Desg FWD 2000000   128.3    Shr Bound(PVST)
Et2/0               Desg FWD 2000000   128.9    Shr
Et2/1               Desg FWD 2000000   128.10   Shr
Et2/2               Desg FWD 2000000   128.11   Shr
Et2/3               Desg FWD 2000000   128.12   Shr
Et3/0               Desg FWD 2000000   128.13   Shr
Et3/1               Desg FWD 2000000   128.14   Shr
 --More--
```

Bmarco11 — D1 — telnet 192.168.71.128 5005 — 80×26

```
Et3/3               Desg FWD 2000000   128.16   Shr

D1#show spanning-tree mst

##### MST0    vlans mapped:   1-4094
Bridge        address aabb.cc00.0100  priority      32768 (32768 sysid 0)
Root          this switch for the CIST
Operational   hello time 2 , forward delay 15, max age 20, txholdcount 6
Configured    hello time 2 , forward delay 15, max age 20, max hops    20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Et0/0            Desg FWD 2000000   128.1    Shr Bound(PVST)
Et0/1            Desg FWD 2000000   128.2    Shr Bound(PVST)
Et0/2            Desg FWD 2000000   128.3    Shr Bound(PVST)
Et2/0            Desg FWD 2000000   128.9    Shr
Et2/1            Desg FWD 2000000   128.10   Shr
Et2/2            Desg FWD 2000000   128.11   Shr
Et2/3            Desg FWD 2000000   128.12   Shr
Et3/0            Desg FWD 2000000   128.13   Shr
Et3/1            Desg FWD 2000000   128.14   Shr
Et3/2            Desg FWD 2000000   128.15   Shr
Et3/3            Desg FWD 2000000   128.16   Shr

D1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5007 — 80×20

```
A1(config-vlan)#name fourthvlan
A1(config-vlan)#exit
A1(config)#vlan 5
A1(config-vlan)#name fithvlan
A1(config-vlan)#name fifthvlan
A1(config-vlan)#exit
A1(config)#end
A1#
*Oct 26 20:16:31.947: %SYS-5-CONFIG_I: Configured from console by console
A1#show spanning-tree root

                                        Root    Hello Max Fwd
Vlan                Root ID             Cost    Time  Age Dly  Root Port
---------------- -------------------- --------- ----- --- ---  ------------
VLAN0001         32768 aabb.cc00.0100       100     2   20  15  Et0/1
VLAN0002         32768 aabb.cc00.0100       100     2   20  15  Et0/1
VLAN0003         32768 aabb.cc00.0100       100     2   20  15  Et0/1
VLAN0004         32768 aabb.cc00.0100       100     2   20  15  Et0/1
VLAN0005         32768 aabb.cc00.0100       100     2   20  15  Et0/1
A1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5005 — 80×26

```
Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Et0/0            Desg FWD 2000000   128.1    Shr Bound(PVST)
Et0/1            Desg FWD 2000000   128.2    Shr Bound(PVST)
Et0/2            Desg FWD 2000000   128.3    Shr Bound(PVST)
Et2/0            Desg FWD 2000000   128.9    Shr
Et2/1            Desg FWD 2000000   128.10   Shr
Et2/2            Desg FWD 2000000   128.11   Shr
Et2/3            Desg FWD 2000000   128.12   Shr
Et3/0            Desg FWD 2000000   128.13   Shr
Et3/1            Desg FWD 2000000   128.14   Shr
Et3/2            Desg FWD 2000000   128.15   Shr
Et3/3            Desg FWD 2000000   128.16   Shr

D1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#spanning-tree mst configuration
D1(config-mst)#name CCNPv8
D1(config-mst)#revision 1
D1(config-mst)#instance 1 vlan 2
D1(config-mst)#instance 2 vlan 4
D1(config-mst)#exit
D1(config)#end
D1#
*Oct 26 20:27:43.999: %SYS-5-CONFIG_I: Configured from console by console
D1#
```

Bmarco11 — D1 — telnet 192.168.71.128 5005 — 80×33

```
D1(config-mst)#exit
D1(config)#end
D1#
*Oct 26 20:27:43.999: %SYS-5-CONFIG_I: Configured from console by console
D1#show spanning-tree mst

##### MST0     vlans mapped:   1,3,5-4094
Bridge         address aabb.cc00.0100  priority      32768 (32768 sysid 0)
Root           this switch for the CIST
Operational    hello time 2 , forward delay 15, max age 20, txholdcount 6
Configured     hello time 2 , forward delay 15, max age 20, max hops    20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Et0/0            Desg FWD 2000000   128.1    Shr Bound(PVST)
Et0/1            Desg FWD 2000000   128.2    Shr
Et0/2            Desg FWD 2000000   128.3    Shr
Et2/0            Desg FWD 2000000   128.9    Shr
Et2/1            Desg FWD 2000000   128.10   Shr
Et2/2            Desg FWD 2000000   128.11   Shr
Et2/3            Desg FWD 2000000   128.12   Shr
Et3/0            Desg FWD 2000000   128.13   Shr
Et3/1            Desg FWD 2000000   128.14   Shr
Et3/2            Desg FWD 2000000   128.15   Shr
Et3/3            Desg FWD 2000000   128.16   Shr

##### MST1     vlans mapped:   2
Bridge         address aabb.cc00.0100  priority      32769 (32768 sysid 1)
Root           this switch for MST1

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
  --More--
```

Bmarco11 — A1 — telnet 192.168.71.128 5007 — 80×33

```
##### MST0     vlans mapped:   1-2,4,6-4094
Bridge         address aabb.cc00.0300  priority      32768 (32768 sysid 0)
Root           address aabb.cc00.0100  priority      32768 (32768 sysid 0)
               port    Et0/2           path cost     1000
Regional Root  this switch
Operational    hello time 2 , forward delay 15, max age 20, txholdcount 6
Configured     hello time 2 , forward delay 15, max age 20, max hops    20

Interface         Role Sts Cost       Prio.Nbr Type
----------------- ---- --- ---------- -------- --------------------------------
Et0/1             Altn BLK 2000000    128.2    Shr Bound(RSTP)
Et0/2             Root FWD 1000       128.3    Shr Bound(RSTP)
Et1/0             Altn BLK 2000000    128.5    Shr Bound(PVST)
Et1/1             Altn BLK 2000000    128.6    Shr Bound(PVST)
Et2/0             Desg LRN 2000000    128.9    Shr
Et2/1             Desg LRN 2000000    128.10   Shr
Et2/2             Desg LRN 2000000    128.11   Shr
Et2/3             Desg LRN 2000000    128.12   Shr
Et3/0             Desg LRN 2000000    128.13   Shr
Et3/1             Desg LRN 2000000    128.14   Shr
Et3/2             Desg LRN 2000000    128.15   Shr
Et3/3             Desg LRN 2000000    128.16   Shr

##### MST1     vlans mapped:   3
Bridge         address aabb.cc00.0300  priority      32769 (32768 sysid 1)
Root           this switch for MST1

Interface         Role Sts Cost       Prio.Nbr Type
----------------- ---- --- ---------- -------- --------------------------------
Et0/1             Altn BLK 2000000    128.2    Shr Bound(RSTP)
Et0/2             Mstr FWD 1000       128.3    Shr Bound(RSTP)
Et1/0             Altn BLK 2000000    128.5    Shr Bound(PVST)
```

Bmarco11 — D1 — telnet 192.168.71.128 5005 — 80×21

```
D1#
D1#show spanning-tree mst configuration digest
Name      [CCNPv8]
Revision  1     Instances configured 3
Digest          0x746D865FEAD726D8F401F9396B8B62DA
Pre-std Digest  0xDE5D7C8B79A99142EBC0A1C265ED7B05
D1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
D1(config)#spanning-tree mst configuration
D1(config-mst)#show current
Current MST configuration
Name      [CCNPv8]
Revision  1     Instances configured 3

Instance  Vlans mapped
--------  -------------------------------------------------------------------
0         1,3,5-4094
1         2
2         4
-------------------------------------------------------------------------------
D1(config-mst)#
```

Bmarco11 — D1 — telnet 192.168.71.128 5005 — 80×21

```
2         4
-------------------------------------------------------------------------------
D1(config-mst)#revision 2
D1(config-mst)#instance 1 vlan 3
D1(config-mst)#instance 2 vlan 5
D1(config-mst)#show pending
Pending MST configuration
Name      [CCNPv8]
Revision  2     Instances configured 3

Instance  Vlans mapped
--------  -------------------------------------------------------------------
0         1,6-4094
1         2-3
2         4-5
-------------------------------------------------------------------------------
D1(config-mst)#exit
D1(config)#end
D1#
*Oct 26 20:35:05.090: %SYS-5-CONFIG_I: Configured from console by console
D1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5007 — 80×26

```
A1#
A1#show spanning-tree mst 1

##### MST1    vlans mapped:   3
Bridge        address aabb.cc00.0300  priority      32769 (32768 sysid 1)
Root          this switch for MST1

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -------------------------------
Et0/1            Altn BLK 2000000   128.2    Shr Bound(RSTP)
Et0/2            Mstr FWD 1000      128.3    Shr Bound(RSTP)
Et1/0            Altn BLK 2000000   128.5    Shr Bound(PVST)
Et1/1            Altn BLK 2000000   128.6    Shr Bound(PVST)

A1#show spanning-tree blockedports

Name                 Blocked Interfaces List
-------------------- ------------------------------------
MST0                 Et0/1, Et1/0, Et1/1
MST1                 Et0/1, Et1/0, Et1/1
MST2                 Et0/1, Et1/0, Et1/1

Number of blocked ports (segments) in the system : 9

A1#
```

Bmarco11 — A1 — telnet 192.168.71.128 5007 — 80×26

```
Et1/1            Altn BLK 2000000   128.6    Shr Bound(PVST)

A1#show spanning-tree blockedports

Name                 Blocked Interfaces List
-------------------- ------------------------------------
MST0                 Et0/1, Et1/0, Et1/1
MST1                 Et0/1, Et1/0, Et1/1
MST2                 Et0/1, Et1/0, Et1/1

Number of blocked ports (segments) in the system : 9

A1#show spanning-tree mst 2

##### MST2    vlans mapped:   5
Bridge        address aabb.cc00.0300  priority      32770 (32768 sysid 2)
Root          this switch for MST2

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -------------------------------
Et0/1            Altn BLK 2000000   128.2    Shr Bound(RSTP)
Et0/2            Mstr FWD 2000000   128.3    Shr Bound(RSTP)
Et1/0            Altn BLK 2000000   128.5    Shr Bound(PVST)
Et1/1            Altn BLK 2000000   128.6    Shr Bound(PVST)

A1#
```